

SQL Assignment Report: Database Management

Student Details

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- **Server Name:** MOUNI\SQLEXPRESS

Task 1: SQL Views Exercise

Description

Developed base tables for Departments and Employees and established relational integrity. Implemented four SQL Views to handle business logic: data retrieval, full name concatenation, annual salary calculations, and a comprehensive employee report with bonus estimation.

SQL Source Code

```
-- 1. Tables creation
DROP TABLE IF EXISTS Employees;
DROP TABLE IF EXISTS Departments;

CREATE TABLE Departments (
    DepartmentID INT PRIMARY KEY,
    DepartmentName VARCHAR(100)
);

CREATE TABLE Employees (
    EmployeeID INT PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    DepartmentID INT FOREIGN KEY REFERENCES Departments(DepartmentID),
    Salary DECIMAL(10, 2),
    JoinDate DATE
);

-- 2. Creating Views
GO
CREATE OR ALTER VIEW vw_EmployeeBasicInfo AS
SELECT e.EmployeeID, e.FirstName, e.LastName, d.DepartmentName
FROM Employees e
JOIN Departments d ON e.DepartmentID = d.DepartmentID;

CREATE OR ALTER VIEW vw_EmployeeReport AS
```

```

SELECT e.EmployeeID, e.FirstName + ' ' + e.LastName AS FullName,
d.DepartmentName,
      (e.Salary * 12) AS AnnualSalary, (e.Salary * 12) * 0.10 AS
Bonus
FROM Employees e
JOIN Departments d ON e.DepartmentID = d.DepartmentID;

```

Execution Output

The following output demonstrates the successfully executed view queries, showcasing formatted employee data including annual salary and bonus projections.

Results Messages						
	EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate
1	101	Rahul	Sharma	1	60000.00	2024-01-15
2	102	Priya	Reddy	2	45000.00	2023-05-10
3	103	Amit	Verma	3	75000.00	2025-02-20
	EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate
1	101	Rahul	Sharma	1	60000.00	2024-01-15
	EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate
1	101	Rahul	Sharma	1	60000.00	2024-01-15
2	103	Amit	Verma	3	75000.00	2025-02-20
	TotalEmployeesInDept1					
1	1					

Query executed successfully. MOUNI\SQLXPRESS (16.0 RTM) MOUNI\MOUNIKAA (56) master 00:00:00 7 rows

Task 2: SQL Stored Procedures Exercise

Description

Designed procedural modules to implement database control logic. Included simple select procedures, parameterized filters for department-based queries, logic for salary/date filtering, and output parameters for aggregate counting.

SQL Source Code

```

-- 1. Simple Stored Procedure
CREATE OR ALTER PROCEDURE sp_GetAllEmployees AS
BEGIN SELECT * FROM Employees; END;

-- 2. Procedure with Input Parameter
CREATE OR ALTER PROCEDURE sp_GetEmployeesByDepartment @DeptID INT AS
BEGIN SELECT * FROM Employees WHERE DepartmentID = @DeptID; END;

-- 3. Procedure with Output Parameter
CREATE OR ALTER PROCEDURE sp_GetEmployeeCountByDept @DeptID INT,
@EmpCount INT OUTPUT AS
BEGIN

```

```

SELECT @EmpCount = COUNT(*) FROM Employees WHERE DepartmentID =
@DeptID;
END;

```

Execution Output

The procedural execution confirms that all filter conditions and output mappings were processed correctly by the SQL Server engine.

The screenshot displays the Microsoft SQL Server Management Studio interface. The main window shows a SQL script with four test cases. The 'Results' pane at the bottom shows the output of the stored procedure for each test case.

Test 1: Simple SP Run

```
EXEC sp_GetAllEmployees;
```

Test 2: Filter by Dept ID 1

```
EXEC sp_GetEmployeesByDepartment @DeptID = 1;
```

Test 3: Multiple inputs test

```
EXEC sp_FilterEmployees @MinSalary = 50000.00, @CutoffDate = '2023-01-01';
```

Test 4: Output parameter test

```
DECLARE @TotalCount INT;
EXEC sp_GetEmployeeCountByDept @DeptID = 1, @EmpCount = @TotalCount OUTPUT;
SELECT @TotalCount AS TotalEmployeesInDept1;
```

Results:

EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate	
1	101	Rahul	Sharma	1	60000.00	2024-01-15
2	102	Priya	Reddy	2	45000.00	2023-05-10
3	103	Amit	Verma	3	75000.00	2025-02-20

EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate	
1	101	Rahul	Sharma	1	60000.00	2024-01-15

EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate	
1	101	Rahul	Sharma	1	60000.00	2024-01-15
2	103	Amit	Verma	3	75000.00	2025-02-20

TotalEmployeesInDept1
1

Query executed successfully.

Conclusion

All database tasks, including relational design, view generation, and stored procedure implementation, were executed successfully. The output confirms that the database engine handles the provided business logic correctly, ensuring data integrity and efficient information retrieval.